



Adult, Failed Airway

Definition of Failed Airway:

Unable to Ventilate and Oxygenate $\geq 90\%$ during
or after one (1) or more unsuccessful intubation attempts

and/ or

Anatomy inconsistent with continued attempts

and/ or

Three (3) unsuccessful attempts by most experienced Paramedic/AEMT.
*Each attempt should include change in approach
or equipment*

NO MORE THAN THREE (3) ATTEMPTS TOTAL

Capnography Monitoring

- End-tidal (EtCO₂) monitoring is mandatory following placement of an endotracheal tube.
- EtCO₂ monitoring is mandatory following placement of a BIAD once available on scene.

Protocols AR 1, 2, and 3 should be utilized together (even if agency is not using Drug Assisted Airway as they contain useful information for airway management.

Failed Airway

Call for additional
resources if available

BVM
Adjunctive Airway NPA/ OPA
Maintains
Oxygen Saturation $\geq 92\%$

YES

Continue BVM
Supplemental Oxygen

Exit to
Appropriate Protocol(s)

NO

B	Attempt Airway Blind Insertion Airway Device Procedure
A	Airway Video Laryngoscopy Device Procedure <i>if available</i> <i>Optional</i>
P	Airway Cricothyrotomy Surgical Procedure
	Supplemental oxygen BVM with Airway Adjuncts Maintain Oxygen Saturation $\geq 92\%$
	Post-intubation BIAD Management Protocol AR 8

 **Notify Destination or
Contact Medical Control** 



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Pearls

- For the purposes of this protocol a secure airway is when the patient is receiving appropriate oxygenation and ventilation.
- If an effective airway is being maintained by BVM with continuous pulse oximetry values of $\geq 90\%$, it is acceptable to continue with basic airway measures.
- Ventilation rate should be 10 - 12 per minute to maintain a EtCO₂ of 35-45 and avoid hyperventilation.
- **Anticipating the Difficult Airway and Airway Assessment**
 - Difficult BVM Ventilation (ROMAN):** Radiation treatment/ Restriction; Obese/ Obstruction/ OB – 2d and 3d trimesters/ Obstructive sleep apnea; Mask seal difficulty (hair, secretions, trauma); Age ≥ 55 ; No teeth.
 - Difficult Laryngoscopy (LEON):** Look externally for anatomical problems; Evaluate 3-3-2 (Mouth opening should equal 3 of patients finger's width, mental area to neck should equal 3 of patient's finger's width, base of chin to thyroid prominence should equal 2 of patients finger's width); Obese, obstruction, OB – 2d and 3d trimesters; Neck mobility limited.
 - Difficulty BIAD (RODS):** Radiation treatment/ Restriction; Obese/ Obstruction/ OB – 2d and 3d trimesters/ Obstructive sleep apnea; Distorted or disrupted airway; Short thyromental distance/ Small mandible.
 - Difficulty Cricothyrotomy / Surgical Airway (SMART):** Surgery scars; Mass or hematoma, Access or anatomical problems; Radiation treatment to face, neck, or chest; Tumor
- Complete an Airway Evaluation Form with any BIAD or Intubation procedure where medications are used to facilitate.
- **Nasotracheal intubation:**
 - Procedure requires spontaneous breathing and may require considerable time, exposing patient to critical desaturation.
 - Contraindicated in combative, anatomically disrupted or distorted airways, increased ICP, severe facial trauma, basal skull fracture, and head injury. Orotracheal route is preferred.
- Intubation attempt defined as laryngoscope blade passing the teeth or endotracheal tube passed into the nostril.
- If First intubation attempt fails, make an adjustment and try again: (Consider change of provider in addition to equipment)
- AEMT and Paramedics should consider using a BIAD if oral-tracheal intubation is unsuccessful.
- During intubation attempts use External Laryngeal Manipulation to improve view of glottis.
- Gastric tube placement should be considered in all intubated patients if available or time allows.
- It is important to secure the endotracheal tube well to better maintain ETT placement. Manual stabilization of endotracheal tube should be used during all patient moves/ transfers.
- **DOPE:** Displaced tracheostomy tube/ ETT, Obstructed tracheostomy tube/ ETT, Pneumothorax and Equipment failure.